

Cristian Palma - Mathematical Engineer

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Education:

MSc in Applied Mathematics on Combinatorial Optimization Algorithms, University of Chile. *Chile, 2022 - 2023*
Thesis on Approximation Algorithms for Concave Load Balancing.
BSc in Mathematics and Computer Science Engineering, École CentralSupélec. *France, 2019 - 2021*
BSc in Common Plan of Engineering, University of Chile. *Chile, 2017 - 2019*

Work Experience:

Research Assistant at Center for Mathematical Modeling *Chile, 2022 - 2023*
Development of Approximation Algorithms for Concave Load Balancing.
Software Engineer Intern at Bigblue *France, 2021*
Problem Solving and Backend dev on Golang with FoundationDB on e-commerce logistics.
Research Intern at Sondra Signal Processing Lab *France, 2020*
Testing an Order Selection method for **Image Processing** and **Finance** on Matlab.
Increased Return on Investment on 50% by using Minimum Variance Portfolio.

Teaching Assistantships:

Graph Theory (2023) and **Calculability and Computational Complexity** (2023) at University of Chile
Mixed Linear Programming (2022, 2023) and **Combinatorial Algorithms** (2022) at University of Chile
Partial Differential Equations (2021), **Measure Theory and Probabilities** (2020) at École CentraleSupélec
Multivariable Calculus (2019) **Algorithms and Data Structures** (2018) at University of Chile
Differential and Integral Calculus (2023, 2022) at University of Chile
Introduction to **Calculus** (2023, 2018) and **Algebra** (2023, 2022) at University of Chile

Projects:

Multipath Transmission Protocols for Load Balancing on Networks dev on **Python**. *2024*
Ranked 15th at IMC Challenge of Huawei.
Insurance Revaluation Optimization using **Clustering** and **Parametric Optimization** on Matlab. *2021*
Noise Control using **Parametric Optimization** of PDE on Python. *2020*
Propagation Maximization Algorithm on Python for a Viral Marketing Campaign. *2020*
Geoproximity based Social Network prototype Full Stack dev for Facebook Hackathon Chile. *2019*

Honors:

Eiffel Excellence Scholarship granted by the Ministry of Europe and Foreign Affairs. *2019 - 2021*
First Place on Engineering Generation Ranking, University of Chile. *2018*
Andrés Bello Excellence Scholarship granted by the University of Chile. *2017 - 2022*
Golden Medal at the National Mathematical Olympiad of SOMACHI, Chile. *2016*

Talks:

XIX Summer School in Discrete Mathematics, Center for Mathematical Modelling. *Chile, 2024*
Discrete Mathematics Session of the XCI Annual Meeting of the SOMACHI. *Chile, 2023*
Algorithms, Game Theory, Combinatorics and Optimization (AGCO) Seminar. *Chile, 2023*
Seminar of Master Students of the Mathematical Engineering Department, University of Chile. *Chile, 2023*
Symposium of Graduate Students of the Engineering Department, University of Chile. *Chile, 2023*

Thesis: Palma C, Soto JA. Load Balancing under Identical Concave Functions.

Computer Skills: Programming: Python, Matlab, JuliaLang, Java, Clang, Golang and MySQL.

Languages: English (Fluent), French (Fluent), German (Elementary) and Spanish (Native).

Free Time: Volleyball, Bodybuilding, Dancing, Running, Kung Fu and **Competitive Programming**.